

Quiz — 2.2.1 Programming Techniques

Name: _____ Date: _ **Mark:** ____ / 25

OCR H446 · Component 02 · 2.2.1

Section A — Multiple choice (8 marks)

A1. Which loop is guaranteed to execute its body **at least once**? A. `while` B. `for...next` C. `do...until` D. `if...else` Your answer: ☐

A2. What is the defining difference between a function and a procedure? A. A function can take parameters; a procedure cannot B. A function returns a single value; a procedure does not return a value C. A procedure is recursive; a function is iterative D. A function is global; a procedure is local Your answer: ☐

A3. A recursive subroutine with no reachable base case will most likely cause: A. A syntax error B. An infinite `while` loop C. A stack overflow D. A logic error that returns 0 Your answer: ☐

A4. When an argument is passed **by reference**, what is passed to the subroutine? A. A copy of the value B. The memory address of the original variable C. The data type only D. A new global variable Your answer: ☐

A5. A variable declared inside a subroutine is: A. Global, visible everywhere B. Local, visible only within that subroutine C. A constant D. Automatically passed by reference Your answer: ☐

A6. Which OOP concept keeps attributes **private** and provides public methods to access them? A. Inheritance B. Polymorphism C. Encapsulation D. Overloading Your answer: ☐

A7. A subclass replaces an inherited method with one of the **same name and parameters**. This is: A. Overloading B. Overriding C. Instantiation D. Encapsulation Your answer: ☐

A8. Which IDE feature lets a programmer **pause execution at a chosen line** to inspect variables? A. Syntax highlighting B. Auto-indentation C. Breakpoint D. Code completion Your answer: ☐

Section B — Short answer (12 marks)

B1. State the **two** parts every correctly written recursive subroutine must contain, and give the role of each. [2]

B2. Consider the program below. State the value printed and explain why, referring to **scope**. [3]

```
total = 10

procedure update()
    total = 5
    print(total)
endprocedure

update()
print(total)
```

Note: in this language an assignment to `total` inside the procedure creates a **new local** variable `total`; it does not change the global.

Output line 1: __ **Output line 2:** ____

Explanation: _____

B3. A subroutine is called with `x = 3`. Explain what the **final value of x** is after each call below, and why. **[3]**

```
procedure addOneByVal(byVal n)
    n = n + 1
endprocedure

procedure addOneByRef(byRef n)
    n = n + 1
endprocedure
```

After `addOneByVal(x)` : `x =` __ **because** _____

After `addOneByRef(x)` : `x =` __ **because** _____

B4. Trace the call `factorial(4)` for the routine below. Show the value returned at each level and the final result. **[2]**

```
function factorial(n)
  if n <= 1 then
    return 1
  else
    return n * factorial(n - 1)
  endif
endfunction
```

B5. Explain **one** advantage of preferring **local** variables over global variables. **[1]**

B6. Define the terms **class** and **object**, making clear how they relate. **[1]**

Section C — Exam-style question (5 marks)

A `BankAccount` class is being written. Complete the OCR-style class so that it:

- has a **private** attribute `balance` initialised to 0 by the constructor,
- has a method `deposit(amount)` that adds `amount` to the balance,
- has a method `getBalance()` that returns the current balance.

```
class BankAccount
  private balance

  public procedure new()
    // (a) initialise balance

  endprocedure

  // (b) deposit method

  // (c) getBalance method

endclass
```

Write your answers for (a), (b) and (c) below. **[5]**

This image shows a blank sheet of white paper with horizontal ruling lines. The top half of the page features solid blue lines, while the bottom half features dashed blue lines. There are five solid lines in the upper section and four dashed lines in the lower section, providing a template for handwriting practice.